

SRI LANKAN DOMESTIC ELECTRICITY TARIFF

How the tariff should work in a smart prepaid electricity platform

Scope

This guide covers the Sri Lankan DOMESTIC BLOCK TARIFF (not the optional Time-of-Use tariff). It explains the official tariff structure, calculation logic, billing-period rules, and a safe way to implement real-time prepaid deductions.

Tariff status

Rates effective from 11 May 2026 and continued by PUCSL from 31 July 2026 until the next revision.

Reference date: 30 September 2026

Key implementation principle

**Choose the tariff structure from TOTAL billing-period consumption first.
Do not treat all rows as one continuous progressive slab table.**

1. Current Domestic Block Tariff

The tariff below is the PUCSL-approved domestic block tariff in force on the reference date. The table is defined for a 30-day billing cycle.

Consumption group	Billing block	Energy rate (LKR/kWh)	Fixed charge (LKR/month)	When this row applies
A: Total 0-60	0-30 kWh	5.00	80	Total usage is 0-30 kWh
A: Total 0-60	31-60 kWh	9.00	210	Total usage is 31-60 kWh
B: Total 61-180	0-60 kWh	14.00	-	First 60 units after total exceeds 60
B: Total 61-180	61-90 kWh	20.00	400	Total usage is 61-90 kWh
B: Total 61-180	91-120 kWh	28.00	1,000	Total usage is 91-120 kWh
B: Total 61-180	121-180 kWh	44.00	1,500	Total usage is 121-180 kWh
C: Total >180	0-180 kWh	32.50	-	First 180 units after total exceeds 180
C: Total >180	Above 180 kWh	100.00	2,500	Total usage is above 180 kWh

Critical interpretation

Groups A, B and C are ALTERNATIVE calculation structures selected according to total consumption. A customer at 80 kWh does not pay 0-30 at 5, 31-60 at 9, and 61-80 at 20. Once total usage is above 60, the first 60 kWh are recalculated at LKR 14/kWh.

What the fixed charge means

- The fixed charge is selected by the final/normalized consumption band for the billing period.
- For Group B, the 0-60 row has no fixed charge; the charge comes from the band containing total usage.
- For Group C (>180), use the LKR 2,500 fixed charge.

Not covered here

Sri Lanka also has an optional Domestic Time-of-Use (ToU) tariff. Keep ToU as a separate tariff plan in the software; do not mix its peak/day/off-peak rates into the block-tariff engine.

2. Tariff Decision Logic

For a standard 30-day billing cycle, calculate total kWh first and then choose exactly one group.

START → Read total consumption for the current billing cycle
Is total consumption $\leq$ 60 kWh?
YES → GROUP A: 0-30 @ 5.00, 31-60 @ 9.00 → Fixed 80 or 210
NO → Is total consumption $\leq$ 180 kWh?
YES → GROUP B: 0-60 @ 14.00, 61-90 @ 20.00, 91-120 @ 28.00, 121-180 @ 44.00
Fixed charge = 400 / 1,000 / 1,500 according to total band
NO → GROUP C: 0-180 @ 32.50, >180 @ 100.00 → Fixed 2,500

Calculation formulas

```

GROUP A (0-30):  bill = kWh × 5.00 + 80
GROUP A (31-60): bill = 30 × 5.00 + (kWh - 30) × 9.00 + 210

GROUP B (61-90): bill = 60 × 14.00 + (kWh - 60) × 20.00 + 400
GROUP B (91-120): bill = 60 × 14.00 + 30 × 20.00 + (kWh - 90) × 28.00 + 1,000
GROUP B (121-180): bill = 60 × 14.00 + 30 × 20.00 + 30 × 28.00 + (kWh - 120) × 44.00 + 1,500

GROUP C (>180):  bill = 180 × 32.50 + (kWh - 180) × 100.00 + 2,500

```

Boundary behavior is intentional

At 61 kWh and 181 kWh the selected tariff structure changes. This can create a large jump in the calculated bill because earlier units are repriced under the new structure and the fixed charge also changes. Your software must handle this explicitly.

3. Worked Examples

These examples are for a 30-day cycle and show the full bill calculation for the block tariff.

Example A - 45 kWh

Selected tariff: Group A (31-60)

Component	Amount (LKR)
30 kWh × 5.00	150
15 kWh × 9.00	135
Fixed charge	210
TOTAL	495

Example B - 80 kWh

Selected tariff: Group B (61-90)

Component	Amount (LKR)
60 kWh × 14.00	840
20 kWh × 20.00	400
Fixed charge	400
TOTAL	1,640

Example C - 150 kWh

Selected tariff: Group B (121-180)

Component	Amount (LKR)
60 kWh × 14.00	840
30 kWh × 20.00	600
30 kWh × 28.00	840
30 kWh × 44.00	1,320
Fixed charge	1,500
TOTAL	5,100

Example D - 190 kWh

Selected tariff: Group C (>180)

Component	Amount (LKR)
180 kWh × 32.50	5,850
10 kWh × 100.00	1,000
Fixed charge	2,500
TOTAL	9,350

Boundary checks for testing

Usage	Expected bill (LKR)	Why it matters
30 kWh	230	Upper edge of band
31 kWh	369	Boundary transition

60 kWh	630	Upper edge of band
61 kWh	1,260	Boundary transition
90 kWh	1,840	Upper edge of band
91 kWh	2,468	Boundary transition
120 kWh	3,280	Upper edge of band
121 kWh	3,824	Boundary transition
180 kWh	6,420	Upper edge of band
181 kWh	8,450	Boundary transition

4. Billing Periods Longer or Shorter Than 30 Days

PUCSL states that domestic consumption blocks are based on a 30-day billing period and the number of units in each block is prorated according to the number of days in the actual billing period. The fixed-charge band is determined using the average consumption per 30 days.

$$\text{adjusted threshold} = \text{standard 30-day threshold} \times (\text{billing_days} / 30)$$

$$\text{Equivalent 30-day usage} = \text{actual_kWh} \times (30 / \text{billing_days})$$

Example: 35-day billing period

The standard thresholds become approximately:

Standard threshold	35-day threshold	Calculation
30 kWh	35 kWh	$30 \times 35 / 30 = 35.0$
60 kWh	70 kWh	$60 \times 35 / 30 = 70.0$
90 kWh	105 kWh	$90 \times 35 / 30 = 105.0$
120 kWh	140 kWh	$120 \times 35 / 30 = 140.0$
180 kWh	210 kWh	$180 \times 35 / 30 = 210.0$

Recommended approach for your prepaid project

Use a fixed 30-day billing cycle in the application if the project controls the cycle. This removes ambiguity for real-time deductions. If you must reproduce CEB/LECO meter-reading periods exactly, store the actual billing dates, prorate the block limits, and reconcile the final charge when the cycle closes.

Do not reset slabs on top-up

A user adding money to the prepaid wallet must NOT reset their tariff consumption. Consumption blocks reset only when a new billing cycle begins. Top-ups affect cash balance; meter readings affect kWh accumulation.

5. Correct Real-Time Prepaid Implementation

The safest method is to calculate the complete tariff liability at the new cumulative usage and deduct only the difference from the previous liability.

```
previous_bill = calculateBill(previous_total_kWh, tariff, cycle)
new_bill      = calculateBill(new_total_kWh, tariff, cycle)
incremental_charge = new_bill - previous_bill
wallet_balance = wallet_balance - incremental_charge
```

Why this matters

Do not simply multiply each new meter delta by the rate shown for the current block. At 61 kWh and 181 kWh, earlier units are repriced because the tariff group changes. Difference-of-total-bill calculation automatically captures that behavior.

Example: crossing 60 → 61 kWh

```
Bill at 60 kWh = 630
Bill at 61 kWh = 1,260
Charge for moving from 60 to 61 kWh = 1,260 - 630 = 630

This is NOT just 1 kWh × 20 = 20.
```

Example: crossing 180 → 181 kWh

```
Bill at 180 kWh = 6,420
Bill at 181 kWh = 8,450
Charge for moving from 180 to 181 kWh = 2,030

This includes repricing + fixed-charge change, not only the LKR 100 unit rate.
```

Recommended meter event flow

- 1. Receive authenticated meter reading with timestamp and meter ID.
- 2. Validate monotonic reading and reject impossible jumps/duplicates.
- 3. Compute cycle usage = current meter reading - cycle-start reading.
- 4. Load the tariff version effective for this billing cycle.
- 5. Compute previous total bill and new total bill using the same tariff engine.
- 6. Deduct the difference from the wallet and create an immutable ledger entry.
- 7. Update dashboard: kWh used, tariff group, amount spent, wallet balance, and forecast days remaining.
- 8. If balance reaches the configured cutoff policy, trigger the approved disconnect/control workflow.

6. Software Design for the Tariff Configurator

Treat tariffs as versioned configuration data. Never hard-code rates inside UI components or multiple services.

Suggested tariff data model

```
TariffVersion
  id, name, category, effectiveFrom, effectiveTo, status

TariffGroup
  id, tariffVersionId, groupCode, totalMinKWh, totalMaxKWh

TariffBlock
  id, tariffGroupId, blockMinKWh, blockMaxKWh, energyRate
  fixedCharge, fixedChargeAppliesWhenTotalMin, fixedChargeAppliesWhenTotalMax
```

Required calculation API

```
calculateElectricityCost({
  consumptionKWh,
  billingStart,
  billingEnd,
  tariffCategory: "DOMESTIC_BLOCK",
  tariffVersion
})

Return:
  tariffGroup, energyCharge, fixedCharge, totalCharge, breakdown[]
```

Admin configurator rules

- Show Group A (0-60), Group B (61-180), and Group C (>180) visually as separate groups.
- Allow rate/fixed-charge editing only through a new tariff version; preserve previous versions for historical bills.
- Validate no gaps or overlaps inside a group and exactly one matching total-consumption group.
- Require effective date and activation confirmation.
- Add a “Test consumption” panel that calculates a sample bill before activation.
- Keep optional ToU tariffs in a separate configuration screen or tariff type.

Suggested UI columns

Group	Total range	Billing block	Rate	Fixed charge	Edit
A	0-60	0-30	5.00	80	Edit
A	0-60	31-60	9.00	210	Edit
B	61-180	0-60	14.00	-	Edit
B	61-180	61-90	20.00	400	Edit
B	61-180	91-120	28.00	1,000	Edit
B	61-180	121-180	44.00	1,500	Edit
C	>180	0-180	32.50	-	Edit
C	>180	>180	100.00	2,500	Edit

7. Validation, Testing and Operational Rules

Minimum automated tests

- Exact boundaries: 0, 1, 30, 31, 60, 61, 90, 91, 120, 121, 180, 181 kWh.
- Random values inside each band and values well above 180 kWh.
- Cycle lengths such as 28, 29, 30, 31 and 35 days if actual-day proration is supported.
- Tariff version switch: historical cycle must keep the tariff effective for that period.
- Duplicate meter event, out-of-order event, meter rollback, and unrealistic spike handling.
- Top-up must never change accumulated kWh or reset tariff group.
- Wallet ledger sum must equal opening balance + top-ups - tariff deductions - other authorized charges.

Production safeguards

- Use decimal/money types, not binary floating point, for LKR values.
- Round only according to the chosen billing policy and do it in one centralized tariff service.
- Keep an audit trail containing meter reading, old usage, new usage, tariff version, calculation breakdown and wallet deduction.
- Require idempotency keys for meter events so a retry cannot charge the user twice.
- Do not let the frontend calculate authoritative billing totals; the backend tariff service must be the source of truth.
- Alert administrators before a tariff version expires or a new official PUCSL decision becomes effective.

Important for September 2026

PUCSL's current End User Tariff Decisions page still points to the May 2026 tariff schedule and the July 2026 decision continuing the existing tariff. Because tariff reviews can occur periodically, verify PUCSL again before deploying or activating a production tariff version.

Implementation checklist

Check	Requirement
☐	One centralized tariff engine
☐	Versioned tariff data
☐	30-day/prorated cycle support
☐	Correct group selection before block calculation
☐	Difference-of-total-bill prepaid deduction
☐	Immutable wallet ledger
☐	Boundary tests
☐	Admin test calculator
☐	Historical tariff preservation
☐	PUCSL update process

8. Official References

The values and billing rules in this guide were checked against Public Utilities Commission of Sri Lanka (PUCSL) sources available on 30 September 2026.

1. PUCSL - Decision on Electricity Tariffs, May 2026 (Annex 2 contains the tariff table; effective 11 May 2026 until the next revision)

https://www.pucsl.gov.lk/wp-content/uploads/2026/05/Full-Final_Decision-on-Electricity-Tariffs-May-2026.pdf

2. PUCSL - Decision on Electricity Tariffs, July 2026 (decision dated 30 July 2026; continues existing end-user tariffs from 31 July 2026 until next revision)

<https://www.pucsl.gov.lk/wp-content/uploads/2026/07/Full-Decision-on-Electricity-Tariffs-July-2026.pdf>

3. PUCSL - End User Tariff Decisions (current tariff decision index)

<https://www.pucsl.gov.lk/end-user-tariff-decisions/>

4. PUCSL - Domestic Electricity Bill Calculator (approved tariff effective from 11 May 2026; notes billing-period block adjustment)

<https://www.pucsl.gov.lk/calculator/>

5. PUCSL - Guideline on Electricity Billing (billing-period and block-proration principles)

<https://www.pucsl.gov.lk/wp-content/uploads/2020/03/Final-Guideline-Billing.pdf>

Engineering note

This document explains tariff logic for software implementation. It is not a substitute for a future PUCSL tariff decision, CEB/LECO billing instruction, or legal/regulatory advice. Store tariffs as effective-dated versions so a new decision can be activated without code changes.

END OF GUIDE